MILITARY SPECIFICATION
INITIATOR, CARTRIDGE ACTUATED, XM98 ASSEMBLY

1. SCOPE

1.1 This specification covers Initiator, Cartridge Actuated, XM98 Assembly, loaded with Cartridge, Initiator, XM201. The assembly will ultimately be used in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-H-25579	- Hose Assembly, Tetrafluoroethylene, High Temperature, Power Plant, Aircraft

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions

DRAWINGS

Munitions Command

F10535456	- Initiator, Cartridge Actuated, XM98, Assembly
C8830984	- Container, Ammunition, Fiber, M426 for Initiators, Cartridge Actuated, M27, XM98 or M99
C8830985	- Box, Packing, Ammunition, for Initiators, Cartridge Actuated, M27, XM98 or M99 in Fiber Container, M426
IEL-10535456	- Index to Inspection Equipment List for Initiator, Cartridge Actuated, XM98, Assembly

FSC 1375

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(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assembles shall comply with the requirements specified on the applicable drawings and referenced specifications.

3.2 Firing pin release.- The firing pin of the assembled initiator, less safety pin and cartridge, shall release from the cocked position when the application of an axial load of 15 to 30 pounds has withdrawn the initiator pin .370 inches, minimum, and .425 inches, maximum.

3.3 Initiator pin security.- Application of an axial load of 200 to 220 pounds to the initiator pin of the assembled initiator shall not withdraw the initiator pin past the safety pin. Load application shall cause no permanent deformation of the safety pin.

3.4 Breakaway torque.- The chamber and cap of the assembled initiator shall withstand a breakaway torque application of 400 inch pounds without loosening or deformation.

3.5 Residual magnetism.- The initiator shall not deflect the indicator of the compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches.

3.6 Functional.

3.6.1 Ballistics.- Initiators loaded with accepted cartridges and temperature conditioned shall produce pressure values specified in Table I. Pressure values shall be produced at the end of 3 feet of aircraft hose (See Figure 2), tetrafluoroethylene, size 4, as specified in MIL-H-25579.

Table I

Temperature	Pressure (PSI)	Ignition Delay Max. (See 6.2)
-65° to -70° F	2500 to 4000	65 milliseconds
65° to 75° F	2600 to 4000	65 milliseconds
200° to 205° F	2750 to 4000	65 milliseconds

3.6.2 Locked shut.- Initiators loaded with accepted cartridges and temperature conditioned at minus 65 degrees F., and plus 205 degrees F., shall withstand the generated pressures with no evidence of breakage or mechanical failure (See 6.1) of exterior components when fired locked shut (See 6.3).

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3.6.3 The initiator shall not fail to fire.

3.6.4 The initiator shall function without mechanical failure. (See 6.1).

3.7 Workmanship.- Requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.7.1 Metal defects.- The part shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.7.2 Burr.- The part shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.7.3 Foreign matter.- The part or assembly shall not contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550.

4.2 Inspection provisions.

4.2.1 Lot.

4.2.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.2.1.2 Size of lot.- A lot shall consist of an identifiable group of assembled units containing metal parts assemblies produced by a homogeneous process and loaded with a single lot of cartridges.

4.2.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and criteria in Table II. The total number of allowable defects per class for either the major or minor classification shall not exceed the acceptance number of Table II. All non-conforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

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Table II

Lot Size	Sample Size	Major Accept	Major Reject	Minor Accept	Minor Reject
1-300	100% Examination	-	-	-	-
301-800	50	0	1	1	2
801-1800	90	1	2	2	3

4.2.2.1 Cartridge, Initiator, XM201, Dwg. D10535460, covering a detail of Dwg. F10535456.

Defect	Method of Inspection
CRITICAL	
1. Mixed cartridges	Visual
2. Metal defective	Visual
3. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Burr	Visual
102. Foreign matter	Visual
MINOR	
201. Case scratched, wrinkled, bulged or dented	Visual
202. Varnish missing or inadequate	Visual

4.2.2.2 Assembly, Dwg. F10535456, less all drawings except B10520412, D10520552, C10520410, C10520411 and MS 19060-3.

CRITICAL	
1. Firing pin protrusion	Gage
2. Steel ball, missing	Gage
3. Striker hole in firing pin housing obstructed	Visual
MAJOR	
101. Burr	Visual
102. Foreign matter	Visual
MINOR	
None defined	

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4.2.2.3 Assembly, Dwg. F10535456, less Drawings D10535460, D10520407, C10520414 and A8359536.

Defect	Method of Inspection
CRITICAL	
1. Firing pin not cocked	Manual and Visual
2. Any part missing	Visual
3. Improper assembly	Visual
MAJOR	
101. Safety pin loose or deformed	Visual
102. Burr	Visual
103. Foreign matter	Visual
MINOR	
None defined	

4.2.2.4 Assembly, Dwg. F10535456, less all drawings except D10520407, C10520414 and D10535460.

CRITICAL	
1. Cartridge not properly seated	Visual
2. Filter missing or insecure	Visual
3. Nylon patch missing or improperly applied	Visual
MAJOR	
101. Burr	Visual
102. Foreign matter	Visual
MINOR	
None defined	

4.2.2.5 Assembly, Dwg. F10535456.

CRITICAL	
1. Cartridge face not seated against base of firing pin housing	X-ray
2. Any part missing	X-ray
3. Improper assembly	X-ray
4. Marking missing, incorrect or illegible	Visual

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Defect	Method of Inspection
MAJOR	
101. Set screw missing or insecure	Visual
102. Burr	Visual
103. Foreign matter	Visual
MINOR	
201. Thread sealing compound, missing from set screws	Visual
202. Shipping cap missing	Visual
203. Protective finish inadequate or defective	Visual

4.2.2.6 Container, Fiber, Dwg. C8830984.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Gap between body and cover, max.	Manual
202. Contents loose	Manual
203. Metal ends loose or distorted	Manual
204. Cut or damaged through all impregnated layers	Visual
205. Marking missing, incorrect or illegible	Visual
206. Tape incomplete or badly wrinkled	Visual
207. Length of tear tab, min.	Visual
208. Color of tape incorrect	Visual

4.2.2.7 Box, Packing, Dwg. C8830985.

CRITICAL	
None defined	
MAJOR	
None defined	

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Defect		Method of Inspection
MINOR		
201.	Contents loose	Manual
202.	Hardware missing, loose, broken or improperly engaged	Manual and Visual
203.	Strapping missing, loose, broken or improperly assembled	Manual and Visual
204.	Contents of box exposed	Visual
205.	Box damaged	Visual
206.	Marking missing, incorrect or illegible	Visual
207.	Number of units incorrect	Visual
208.	Car seal missing, unsealed or improperly positioned	Visual

4.2.3 Inspection testing.- The following tests shall be performed in accordance with 4.3.

4.2.3.1 Firing pin release.- The initiator pin of each subassembly consisting of all parts except the safety pin, cartridge and chamber, shall be subjected to the firing pin release test. Failure of any subassembly to comply with the requirements of 3.2 shall be cause for rejection of the subassembly.

4.2.3.2 Initiator pin security.- The initiator pin of each subassembly consisting of all parts except the chamber and cartridge, shall be subjected to the initiator pin security test. Failure of any subassembly to comply with the requirements of 3.3 shall be cause for rejection of the subassembly.

4.2.3.3 Breakaway torque.- Each initiator shall be subjected to the breakaway torque application. Failure of any initiator assembly to comply with the requirements of 3.4 shall be cause for rejection of the assembly.

4.2.3.4 Residual magnetism.- Each initiator shall be subjected to the test for excessive residual magnetism. Failure of any initiator to comply with the requirement of 3.5 shall be cause for rejection of the initiator.

4.2.3.5 Functional.- The following tests shall be performed by the Government.

4.2.3.5.1 Ballistics.- Thirty units shall be tested to determine compliance with the requirement of 3.6.1, 3.6.3 and 3.6.4. The units shall be temperature conditioned as provided in Table III and 4.3.5.1.1. Failure of any unit to comply with the requirements of 3.6.1, 3.6.3 and 3.6.4 shall be cause for rejection of the lot.

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Table III

Temperature	Sample Size
-65° to -70° F	10
65° to 75° F	10
200° to 205° F	10

4.2.3.5.2 Locked shut.- Two units from each production lot shall be fired locked shut (See 6.3). Failure of any initiator to comply with the requirements of 3.6.2, 3.6.3 and 3.6.4 shall be cause for rejection of the lot.

4.2.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL-10535456.

4.3 Test methods and procedures.

4.3.1 Firing pin release.- The subassembly shall be secured in the fixture. A pull shall be applied to the initiator pin in the longitudinal direction, in conjunction with the fixture which registers the amount of pull.

4.3.2 Initiator pin security.- The subassembly shall be secured in the fixture with the safety pin in place. A pull shall be applied to the initiator pin in the longitudinal direction in conjunction with a fixture which registers the amount of pull. The test shall be performed with the safety pin in both positions.

4.3.3 Breakaway torque.- A torque wrench set at the torque requirements specified in 3.4 shall be used in conjunction with the fixture to determine the adequacy of the joint. The torque shall be applied to the cap.

4.3.4 Residual magnetism.- The initiator shall be tested for excessive residual magnetism, in an area free of local magnetic effects, by placing the initiator 6 inches from and at the same height as the compass, in a north-south horizontal position (cap end at south), with the compass set in an east-west heading as shown in Figure 1. The initiator shall be moved slowly past the compass in a longitudinal direction, for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the initiator.

4.3.5 Functional.

4.3.5.1 Ballistics.- The test sample shall be rigidly supported in the fixture. The sample shall be tested within 5 minutes after removal from the conditioning chamber (See Figure 2). When this time limit is exceeded, the sample unit shall be reconditioned at the original temperature.

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4.3.5.1.1 Temperature conditioning.- A standard XM98 Initiator, loaded with a dummy XM201 cartridge, containing a temperature sensitive element, shall be placed in the conditioning chamber at the same time as the samples. The conditioning time shall be a minimum of 1 hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

4.3.5.2 Locked shut.- Immediately upon receipt of a lot of metal parts assemblies, the inspector shall select 2 initiators at random from the lot. These samples shall be loaded with cartridges from an accepted lot. Samples shall be temperature conditioned in accordance with 4.3.5.1.1 as follows: one unit at 200 degrees F., to 205 degrees F., and the other unit at minus 65 degrees to minus 70 degrees F. These units shall then be fired locked shut.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.- Level A.- Packing and marking shall be in accordance with Drawings C8830984 and C8830985.

5.2 Data card.- Data card information shall be as specified in MIL-A-2550.

6. NOTES

6.1 A mechanical failure is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.2 Ignition delay is defined as the time that elapses between the firing of the pressure element and the beginning of the continuous pressure rise on the pressure time record.

6.3 Locked shut is defined as a condition which prevents the generated gas pressure from leaving the chamber of the initiator.

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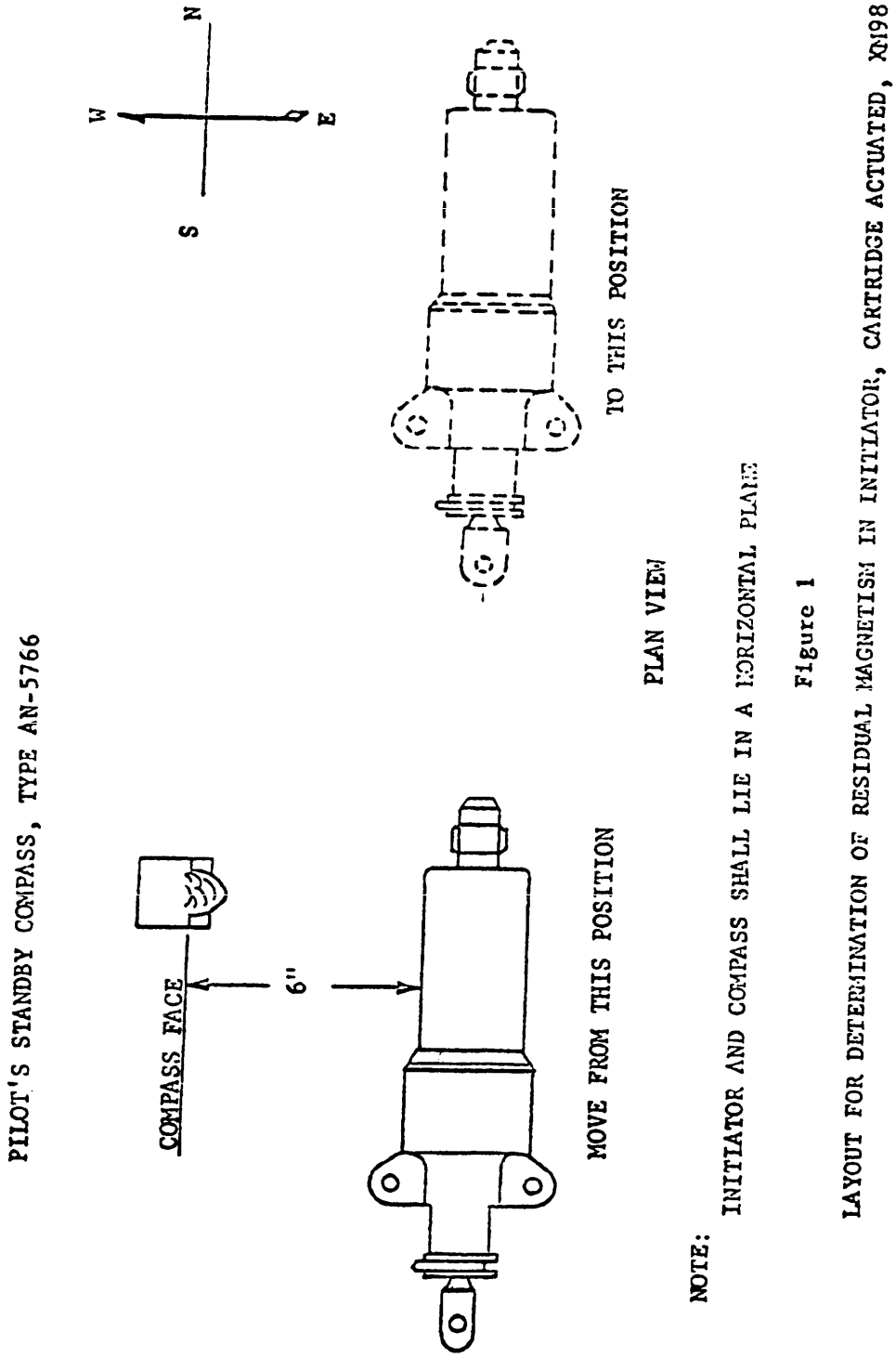
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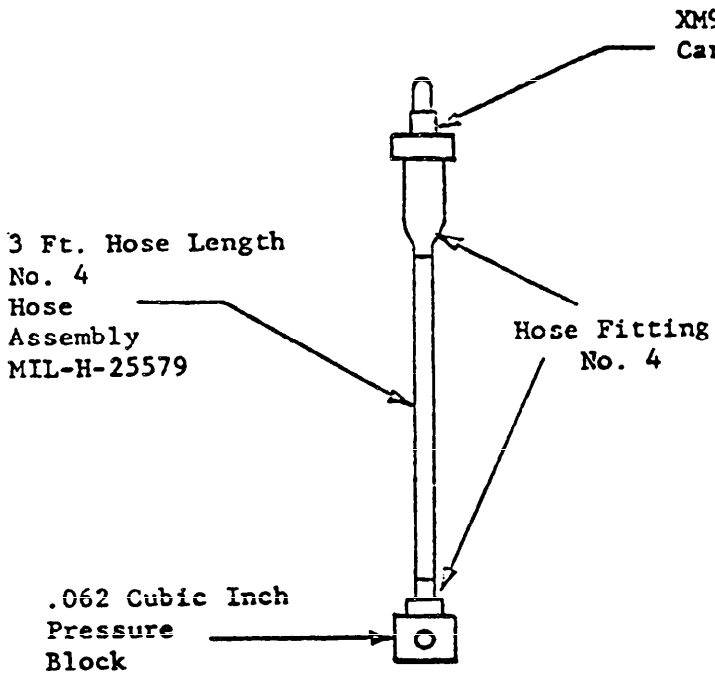
NOTE:
INITIATOR AND COMPASS SHALL LIE IN A HORIZONTAL PLANE

Figure 1

LAYOUT FOR DETERMINATION OF RESIDUAL MAGNETISM IN INITIATOR, CARTRIDGE ACTUATED, XM98

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Name	Format	Description



PERFORMANCE TEST LAYOUT SKETCH

TOP VIEW

Figure 2

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE	
CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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